

Q1 :- Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

Answer :- The difference between a **Tableau Dashboard** and a **Tableau Story Point** lies in their purpose and how they present data. A dashboard is designed for interactive monitoring and exploration, while a story point is designed for narrative storytelling.

Key Differences

Feature	Tableau Dashboard	Tableau Story Point (Tableau Story)
Structure	A single page containing multiple worksheets and objects (charts, maps, filters).	A sequence of individual "points" or slides, each containing a sheet or dashboard.
Interactivity	High. Actions on one chart (like filtering) usually update the other charts on the same dashboard.	Narrative-focused. Each point has a specific "caption" or description to guide the user.
Purpose	Monitoring: Used to track real-time KPIs and allow users to drill down into specific data segments.	Storytelling: Used to walk an audience through a discovery, explain a trend, or support a specific conclusion.
User Experience	Exploratory; the user chooses what to look at and filter.	Guided; the author controls the sequence and context of the data presentation.

Q2 :- What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

Answer :- In Tableau, **Actions** are interactive features that allow users to engage with data in a dynamic way. They add a layer of interactivity to dashboards and worksheets, enabling a user to perform an action (like clicking, hovering, or selecting) on one part of the data to trigger a response in another part.

There are several types of actions in Tableau:

1. **Filter Actions:** Use data from one view to filter data in another view.
 2. **Highlight Actions:** Highlight specific data points across multiple views while keeping all data visible.
 3. **Go to URL Actions:** Open a web page, file, or another Tableau dashboard based on a selection.
 4. **Go to Sheet Actions:** Navigate directly to another worksheet or dashboard.
 5. **Parameter/Set Actions:** Allow users to change parameter values or set memberships by interacting with the marks on a viz.
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How Filter Actions Help Hospital Management (Example: `Hospital_Management.csv`)

A **Filter Action** is particularly powerful for hospital management because it allows for "drill-down" analysis. Instead of looking at a static report, management can interact with a dashboard to find the root cause of issues like high costs or poor outcomes.

Using the provided dataset, here is how a Filter Action would help:

1. Regional Performance Analysis

- **The Scenario:** A dashboard has a Map showing **Hospital Regions** (West, South, Central, etc.) and a Bar Chart showing **Average Treatment Cost**.
- **The Action:** By setting up a Filter Action, a manager can click on the "West" region on the map.
- **The Benefit:** The Bar Chart instantly updates to show costs *only* for the West region. This helps management identify if specific regions are exceeding budgets compared to others.

2. Investigating Patient Outcomes by Diagnosis

- **The Scenario:** One chart shows a count of **Outcomes** (Recovered vs. Critical), and another shows **Doctor Performance**.
- **The Action:** A manager clicks on the "Critical" segment in the outcome chart.
- **The Benefit:** The Doctor chart filters to show which doctors have the highest number of critical cases. This allows management to see if certain doctors (like `Dr. Mehta` or `Dr. Sharma`) are handling more complex cases or if there is a training need in a specific department.

3. Resource Allocation based on Department

- **The Scenario:** A Treemap shows different **Departments** (Cardiology, Neurology, etc.), and a line graph shows **Length of Stay** over time.
- **The Action:** Clicking on "Cardiology" in the treemap filters the line graph.
- **The Benefit:** Management can see if Cardiology patients are staying longer than expected. If **Length_of_Stay** is high, they can investigate if it's due to the **Treatment_Type** (e.g., Surgery vs. Medication) or specific diagnoses like **Heart Disease**.

Q10 :- Based on your visuals, answer :

1. Which department generates the highest treatment cost?
2. Which region has more critical patients?
3. What business decisions can hospital management take from these insights?

Answer :- 1. The **Neurology** department generates the highest treatment cost

2. The **South** and **West** regions are tied for the highest number of critical patients, with each region reporting **20 patients** in critical condition. They are followed by the North (16), Central (15), and East (13) regions.

3. **Specialized Resource Investment:** Since Neurology is the highest revenue-generating and cost-intensive department, management should prioritize the procurement of advanced neurological diagnostic equipment and high-end medical supplies for this unit.

Emergency Capacity Expansion: Management should focus on expanding ICU and emergency care facilities specifically in the **South and West regions**, as these areas show a higher demand for critical care services.

Staffing Optimization: High-priority staffing (e.g., specialized critical care nurses and senior consultants) should be strategically deployed to the South and West regions to maintain high standards of patient safety in those critical zones.

Cost Management: A financial audit could be conducted in the Neurology department to identify if the high costs are due to medical complexities or potential operational inefficiencies that could be optimized to improve profit margins.